M1.1: Chisel Language Hardware Modeling

Theory: Generates structural RTL circuits via Scala-embedded object-oriented abstractions.

Details: Chisel compiles hardware builders into FIRRTL intermediate trees. It represents physical wires, registers, and ports using typed Scala values.

Trade-off: Chisel is structural RTL, not software HLS. Designers must map syntax constructs directly to physical gate-level structures.

M1.2: Configuration & Parameterization System

Theory: Dynamically builds sub-systems using Scala Traits (Cake Pattern) and implicit parameters context.

Details: Passes global parameters (e.g. cache configurations, CPU features) down module trees via implicit Parameter objects. Builds subsystems by mixing Traits.

Trade-off: Highly complex Scala type systems; parameter collisions (e.g., overlapping address maps) cause compile-time aborts rather than runtime faults.

M1.3: Rocket Core 5-Stage Pipeline

Theory: In-order execution featuring IF/ID/EX/MEM/WB stages and bypass logic.

Details: Implements an in-order pipeline, resolving RAW dependencies using EX-to-ID bypassing.

Trade-off: Simple, low-power design, but stalls on long-latency events (e.g., FPU or Cache Misses).

M1.4: FPU & MMU Virtual Memory Translation

Theory: IEEE-754 hardware floating-point and SV39 Page Table Walker with TLBs.

Details: Implements SV39 virtual memory translation featuring private ITLBs/DTLBs and hardware page table walkers that traverse memory pages on TLB misses.

Trade-off: Hardware page walking accelerates translation but increases core size and control logic complexity.

M1.5: Instruction Cache & ITim

Theory: Single-port RAM instruction cache with prefetching and tightly-integrated memory.

Details: Employs single-port SRAM with prefetch buffers for zero-latency instruction dispatch. ITim maps cache regions to physical scratchpads for deterministic timing.

Trade-off: Single-port RAM is compact but halts CPU fetches during cache line refilling.

M1.6: Data Cache & MSHR Architecture

Theory: Non-blocking HellaCache using Miss Status Holding Registers (MSHR).

Details: HellaCache handles D-Cache misses by recording pending transactions in MSHRs and issuing bus requests, allowing independent CPU execution.

Trade-off: MSHRs increase instruction throughput but require complex state tracking logic and consume extra area.

M2.1: TileLink Five-Channel Model

Theory: Decoupled physical channels A, B, C, D, E for request/response and coherency.

Details: Channel A handles Master requests, B tracks Slave probes, C routes L1 releases/writebacks, D transmits Slave responses, and E acknowledges Grants.

Trade-off: Eliminates cyclic channel dependencies to prevent deadlocks, but increases physical routing overhead.

M2.2: MSI Cache Coherency Controller

Theory: L1 MSI state machine transitions driven by L2 probe/release routing.

Details: When a core requests write access (Exclusive), the L2 controller broadcasts Probes (channel B) to other cores, which downgrade and release lines (channel C).

Trade-off: Guarantees cache consistency across cores but introduces massive probe bus overhead under high contention.

M2.3: SystemBus Subsystem Interconnect

Theory: Diplomacy-negotiated subsystem bus routing topologies.

Details: Subsystem networks (SystemBus, PeripheralBus, ControlBus) are mapped using Diplomacy nodes, compiling into optimized bus decoders.

Trade-off: Automates correct connectivity but adding bus hierarchies increases latency for peripheral I/O access.

M2.4: Bus Adapters & Cross-Clock Bridges

Theory: Protocol conversion, width adjustment, and asynchronous boundary crossings.

Details: Translates internal TileLink signals to AXI4/APB formats. Manages width widgets and clock domain crossings using asynchronous synchronization registers.

Trade-off: Bridges provide broad IP compatibility but introduce 2 4 clock cycles of synchronization delay.

M2.5: Crossbars & Bus Arbitration

Theory: Address decoders and multiplexer arbiters with deadlock-free queuing.

Details: TLXbar generates bus crossbars containing address decoders and priority arbiters (e.g. Round-Robin), using FIFO buffers to prevent blocking.

Trade-off: Arbitration choices impact core latency. Large crossbars complicate physical wire routing in backend layouts.

M2.6: Memory Mapped I/O (MMIO) Registers

Theory: Automatically generates address decoding and register maps.

Details: Peripherals implementing ‘RegMapper‘ register ranges during Diplomacy parameter negotiation, compiling into collision-free address decoders.

Trade-off: Address alterations require rebuilding Verilog from Scala configs.

M3.1: JTAG Debug Module

Theory: Implements RISC-V Debug spec 0.13 via JTAG and DMI.

Details: Resolves JTAG shift streams into Debug Module Interface (DMI) register read/write commands, allowing debuggers to halt cores.

Trade-off: Debug modules are crucial for physical testing but must be physically fused or locked to prevent hardware backdoors.

M3.2: BootROM & Reset Logic

Theory: Fixed reset vector execution and devicetree blob (DTB) setup.

Details: On reset, the PC jumps to the BootROM. BootROM code passes the DTB to memory and invokes OpenSBI.

Trade-off: Hardcoded reset vectors are robust but lock boot sequences; upgrades must rely on secondary software bootloaders.

M3.3: PLIC & CLINT Interrupt Controllers

Theory: Aggregates external interrupts and manages core timer ticks.

Details: CLINT handles timer/software interrupts. PLIC routes external signals through an arbitration tree based on priority thresholds.

Trade-off: Centralized PLIC is easy to route, but multi-tile systems face routing congestion, requiring decentralized CLIC approaches.

M3.4: FIRRTL Compiler Infrastructure

Theory: Lowers Chisel AST structures to low-level Verilog code.

Details: FIRRTL compiles high-level Chisel trees through High, Middle, and Low compiler passes, applying optimizations before emitting Verilog.

Trade-off: Multi-pass lowering optimizes netlists but lengthens compile times for large SoC designs.

M3.5: Verilog Codegen & Simulation Harness

Theory: Emits synthesis-ready Verilog and compiles Harness programs.

Details: Generates synthesizable Verilog. Verilator translates this to C++ classes, and the simulation harness injects software binaries into memory blocks.

Trade-off: Speeds up verification but software simulation runs 6 7 orders of magnitude slower than physical hardware.

M3.6: RoCC Coprocessor Interface

Theory: Integrates custom accelerators using custom0-3 opcodes.

Details: Under custom0-3 opcodes, the ID stage dispatches commands and registers to the RoCC bus, halting execution until the coprocessor returns data.

Trade-off: Simple accelerator integration, but coprocessors share L1 D$ and MMU resources, making debugging difficult if conflicts occur.

M3.7: Physical Memory Protection (PMP)

Theory: Enforces physical address access checks at the memory interface.

Details: Compares address registers during the MEM stage, raising exceptions (Access Faults) on unauthorized reads, writes, or executes.

Trade-off: Critical for security but parallel address comparators add combinational logic delay.

M4.1: Card 20 Fallback Title

Placeholder text for compiler robustness.

M4.2: Card 21 Fallback Title

Placeholder text for compiler robustness.

M4.3: Card 22 Fallback Title

Placeholder text for compiler robustness.

M4.4: Card 23 Fallback Title

Placeholder text for compiler robustness.

M4.5: Card 24 Fallback Title

Placeholder text for compiler robustness.

M5.1: Card 25 Fallback Title

Placeholder text for compiler robustness.

M5.2: Card 26 Fallback Title

Placeholder text for compiler robustness.

M5.3: Card 27 Fallback Title

Placeholder text for compiler robustness.

M5.4: Card 28 Fallback Title

Placeholder text for compiler robustness.

Zone T1: Rocket Chip Core APIs

freechips::rocketchip::rocket::RocketCore,

freechips::rocketchip::tilelink::TLBundle,

freechips::rocketchip::subsystem::BaseSubsystem,

freechips::rocketchip::config::Parameters,

freechips::rocketchip::tile::LazyRoCC,

freechips::rocketchip::diplomacy::LazyModule,

freechips::rocketchip::devices::tilelink::PLIC

Zone T2: Compilation & Execution Errors

tilelink_protocol_deadlock, pipeline_hazard_data_stall, cache_coherency_mismatch, pmp_access_violation, firrtl_lowering_compilation_error, debug_module_jtag_timeout